

TUSHAR MISHRA

◇ Bhopal, India - open to remote/relocate
tm23msg@gmail.com ◇ [X](#) ◇ [Portfolio](#) ◇ [Github](#)

EDUCATION

B.Tech in Computer Science, Bhabha eng. research institute, Bhopal
SGPA : 8.21/10

2023 - 2027
MP, India

SKILLS

Programming Languages	TypeScript, Python, HTML, CSS
Frameworks/Libraries	PyTorch, TensorFlow, Hugging Face, Scikit-learn, Next.js, Tailwind
RL/LLM Training	TRL, Prime-RL, Unsloth, OpenEnv, PEFT/LoRA
Agentic AI / VectorDB	LangChain, LangGraph, DSPy, Chroma, Pinecone
Dev Tools/Platforms	Git/GitHub, Docker, FastAPI, WandB

EXPERIENCE

Research Intern - Osaka University, Japan

Present (*Remote*)

- Performing experiments on transfer learning, multi task learning and reading lots of papers. Personally working on RL rust reasoning paper as well.

PROJECTS/OPEN-SOURCE

Social Network - OpenEnv ([Demo](#), [Github](#))

Hugging Face, Synthetic Dataset, LLM Reasoning

- A synthetic social network where a detection agent learns to find coordinated inauthentic behavior — created synthetic data pool of 90K samples and **14K for SFT**.

Rust Algo Reasoning - OpenEnv ([Demo](#), [Github](#))

Rust, Hugging Face, LLM Reasoning, Docker

- Developed for the Meta OpenEnv hackathon; curated a dataset of **2.5k coding problems** featuring algorithmic reasoning, complexity analysis, and a robust **code execution environment**.

Image processor - Multiple domains in one ([Demo](#), [Github](#))

CLIP, Pinecone, YOLOv9, Flan-T5, PyTorch

- Developed an AI-powered RAG-based project covering domains such as **text generation, image captioning, segmentation, and object detection**.

Orbit AI - Exoplanet detection ([Demo](#), [Github](#))

Voting Classifier, CatBoost, Next.js, TypeScript, PyTorch

- Web-based platform built for the NASA Space Apps Challenge to accelerate **exoplanet discovery**; implemented full pipeline including **Preprocessing, Feature Engineering, and Model Architecture** on Kepler/TESS datasets.

Tiny projects ([Github](#))

Gists, TensorFlow, Pytorch

- Currently exploring Fokker planck - Diffusion model papers with deep diving into maths and also we have Built GPT, Llama and MOE Model from scratch with **all novel components** and gpt was trained on 200M tokens of FineWeb dataset. ([Github](#))

EXTRA-CURRICULAR ACTIVITIES

- Actively write [blog posts](#) and Twitter posts ([X](#)) to help people who're starting out in ml and also sharing my journey.